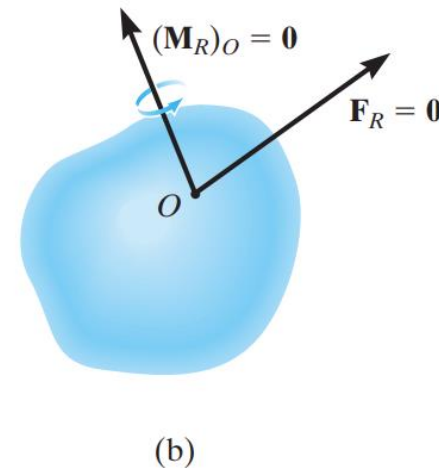
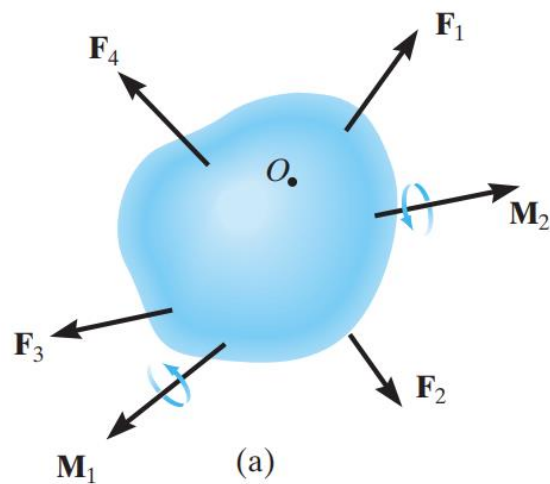


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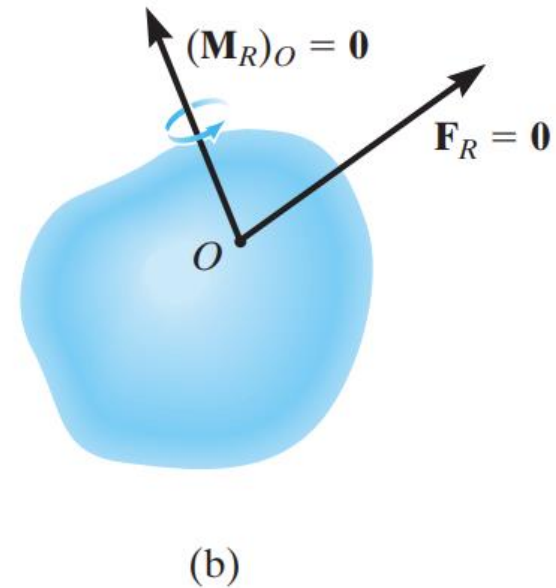
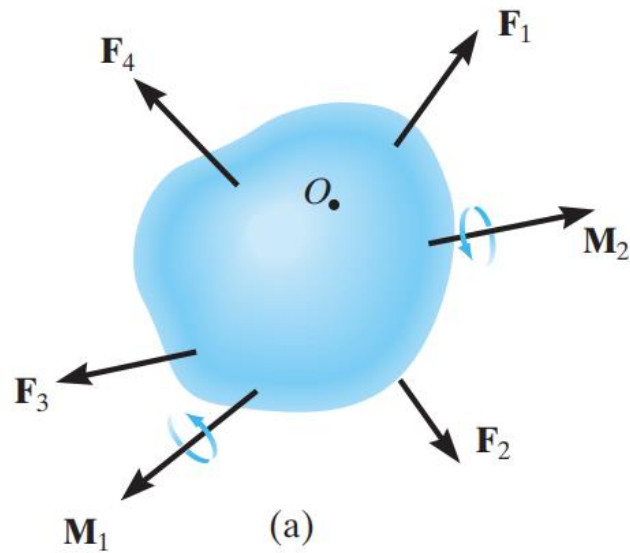
Statics (ME 231)
Lecture 7: Equilibrium of
Rigid Body in 2D
By Saeed Lotfan

Conditions For Rigid-Body Equilibrium

Generally, a rigid body is subjected to an external force and couple moment system that is the result of the effects of gravitational, electrical, magnetic, or contact forces caused by supports or adjacent bodies. The internal forces caused by interactions between particles within the body are not shown below, because these forces occur in equal but opposite collinear pairs and hence will cancel out, a consequence of Newton's third law.



Conditions For Rigid-Body Equilibrium



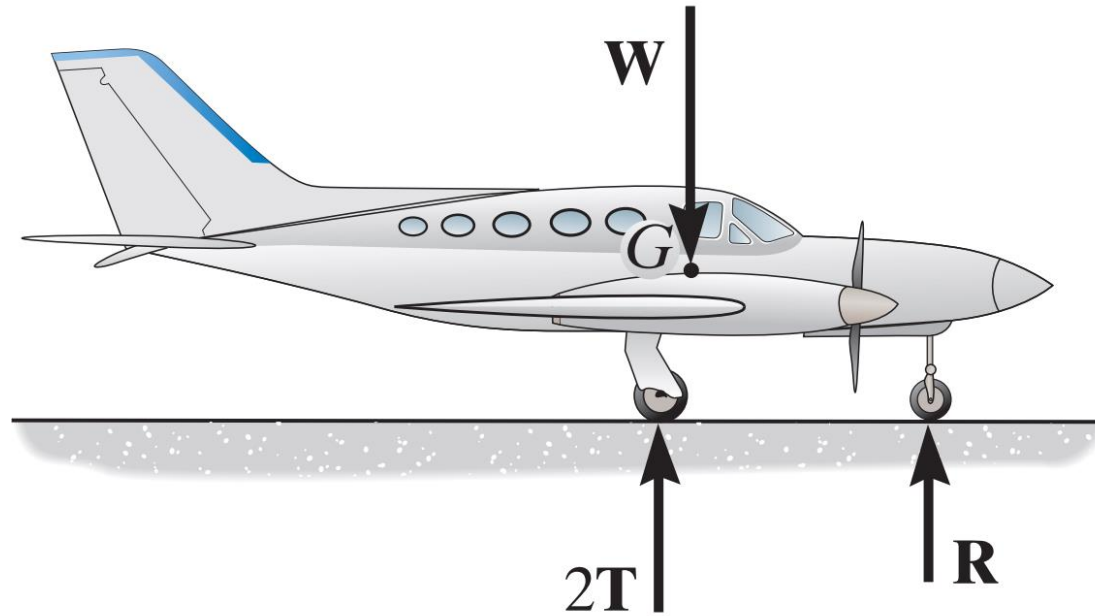
$$\mathbf{F}_R = \Sigma \mathbf{F} = \mathbf{0}$$

$$(\mathbf{M}_R)_O = \Sigma \mathbf{M}_O = \mathbf{0}$$

Conditions For Rigid-Body Equilibrium

Note: When applying the equations of equilibrium, we will assume that the body remains **rigid**. In reality, all bodies deform when subjected to loads; however, most engineering materials such as **steel** and **concrete** are very stiff and so their deformation is usually very small. Therefore, when applying the equations of equilibrium, we can generally assume that the body will remain *rigid* and *not deform* under the applied load without introducing any significant error. This way the direction of the applied forces and their moment arms with respect to a fixed reference remain the same both before and after a load is applied.

Equilibrium in Two Dimensional or *Coplanar* Force System

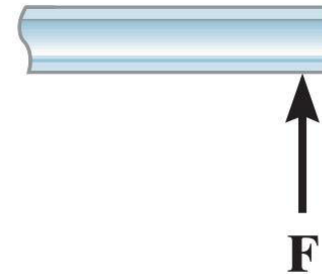
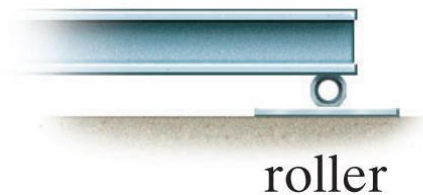


Equilibrium in 2D: Free-Body Diagrams

This diagram is a sketch of the outlined shape of the body, which represents it as being isolated or “free” from its surroundings, i.e., a “free body.”

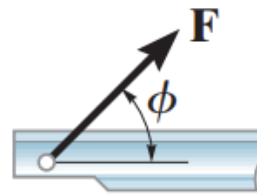
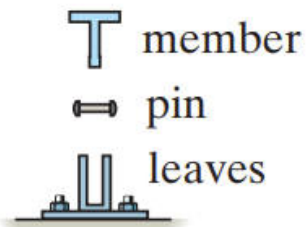
Support Reactions

- A support prevents the translation of a body by exerting a force on the body.
- A support prevents the rotation of a body by exerting a couple moment on the body.

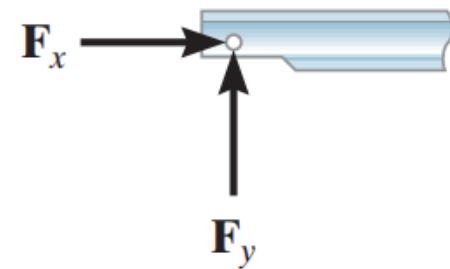


Equilibrium in 2D: Free-Body Diagrams

Support Reactions



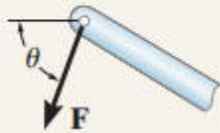
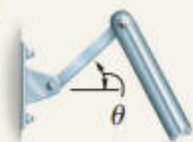
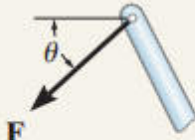
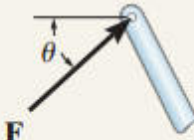
or



Equilibrium in 2D: Free-Body Diagrams

Support Reactions

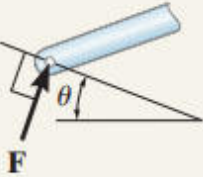
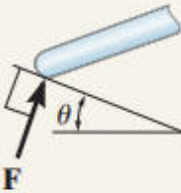
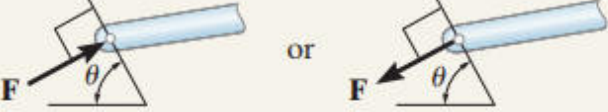
TABLE 5–1 Supports for Rigid Bodies Subjected to Two-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(1)  cable		One unknown. The reaction is a tension force which acts away from the member in the direction of the cable.
(2)  weightless link	 or 	One unknown. The reaction is a force which acts along the axis of the link.
(3)  roller		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.

Equilibrium in 2D: Free-Body Diagrams

Support Reactions

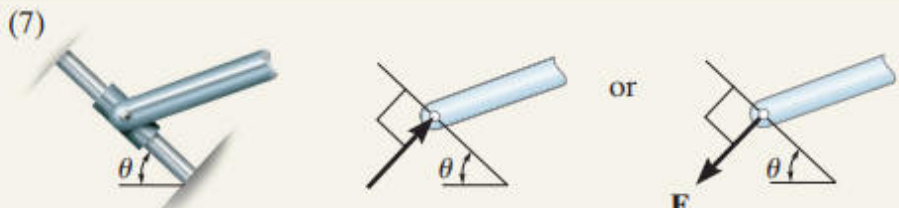
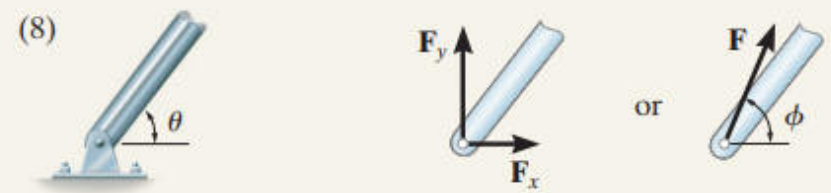
TABLE 5–1 Supports for Rigid Bodies Subjected to Two-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(4)  rocker		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(5)  smooth contacting surface		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(6)  roller or pin in confined smooth slot		One unknown. The reaction is a force which acts perpendicular to the slot.

Equilibrium in 2D: Free-Body Diagrams

Support Reactions

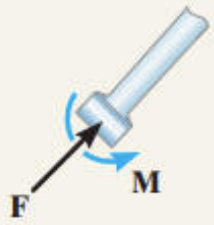
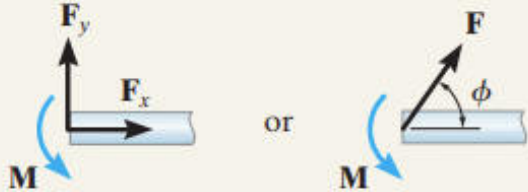
TABLE 5–1 Supports for Rigid Bodies Subjected to Two-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(7)  member pin connected to collar on smooth rod	One unknown. The reaction is a force which acts perpendicular to the rod.	
(8)  smooth pin or hinge	Two unknowns. The reactions are two components of force, or the magnitude and direction ϕ of the resultant force. Note that ϕ and θ are not necessarily equal [usually not, unless the rod shown is a link as in (2)].	

Equilibrium in 2D: Free-Body Diagrams

Support Reactions

TABLE 5–1 Supports for Rigid Bodies Subjected to Two-Dimensional Force Systems

Types of Connection	Reaction	Number of Unknowns
(9)  member fixed connected to collar on smooth rod		Two unknowns. The reactions are the couple moment and the force which acts perpendicular to the rod.
(10)  fixed support		Three unknowns. The reactions are the couple moment and the two force components, or the couple moment and the magnitude and direction ϕ of the resultant force.

Equilibrium in 2D: Free-Body Diagrams

Support Reactions



The cable exerts a force on the bracket in the direction of the cable. (1)



The rocker support for this bridge girder allows horizontal movement so the bridge is free to expand and contract due to a change in temperature. (4)

This concrete girder rests on the ledge that is assumed to act as a smooth contacting surface. (5)



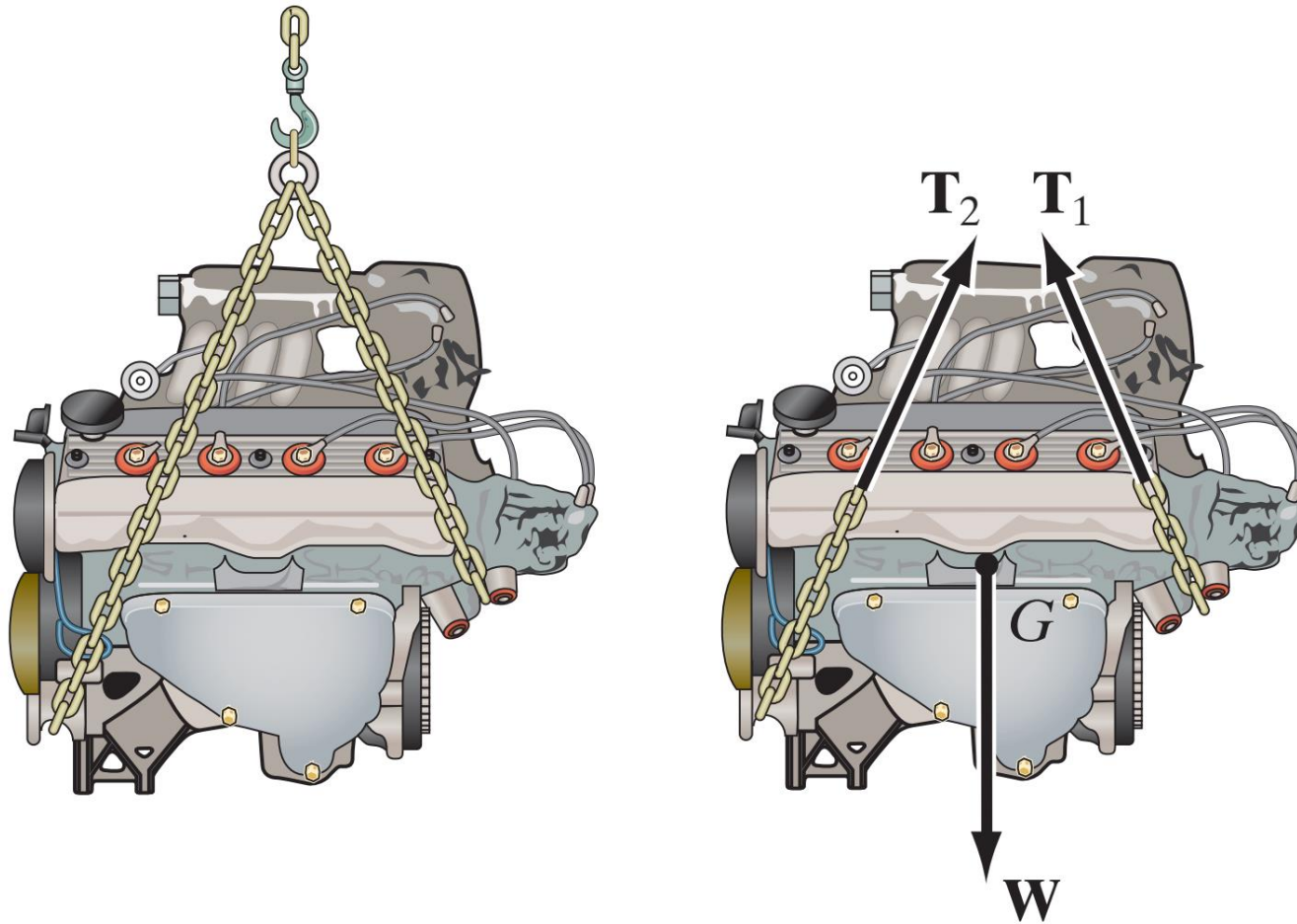
Typical pin support for a beam. (8)

The floor beams of this building are welded together and thus form fixed connections. (10)



Equilibrium in 2D: Free-Body Diagrams

Internal Forces



Equilibrium in 2D: Free-Body Diagrams

Weight and the Center of Gravity

When a body is within a gravitational field, then **each of its particles** has a specified weight. It was shown in distributed loading topic that such a system of forces can be reduced to a single resultant force acting through a specified point. We refer to this force resultant as the weight **W** of the body and to the location of its point of application as the **center of gravity**.

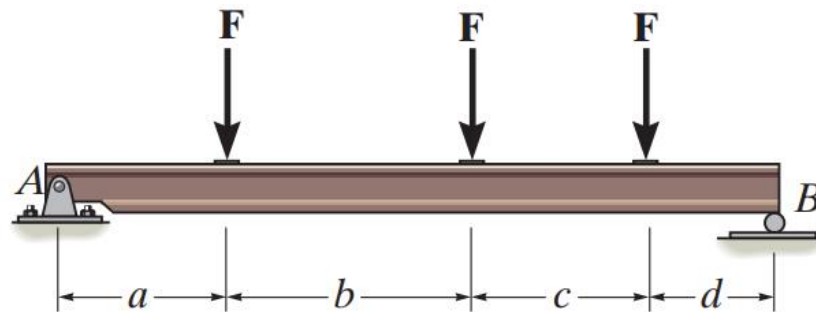
Also, when the body is uniform or made from the same material, the center of gravity will be located at the body's geometric center or centroid; however, if the body consists of a nonuniform distribution of material, or has an unusual shape, then the location of its center of gravity G will be given.

Equilibrium in 2D: Free-Body Diagrams

Idealized Models



(a)



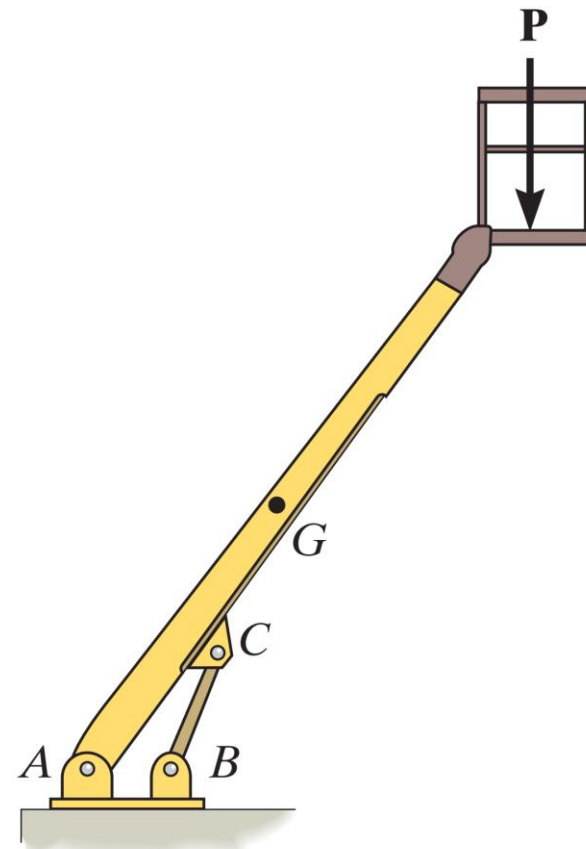
(b)

Equilibrium in 2D: Free-Body Diagrams

Idealized Models



(a)

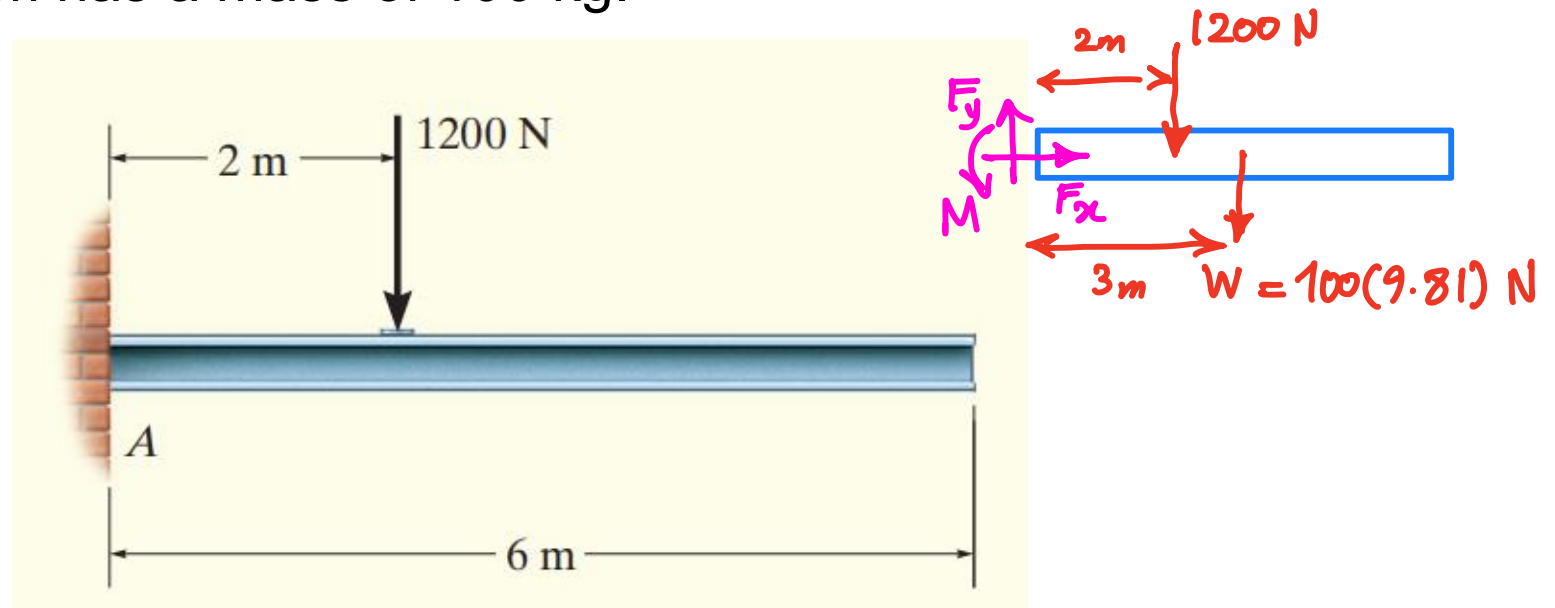


(b)

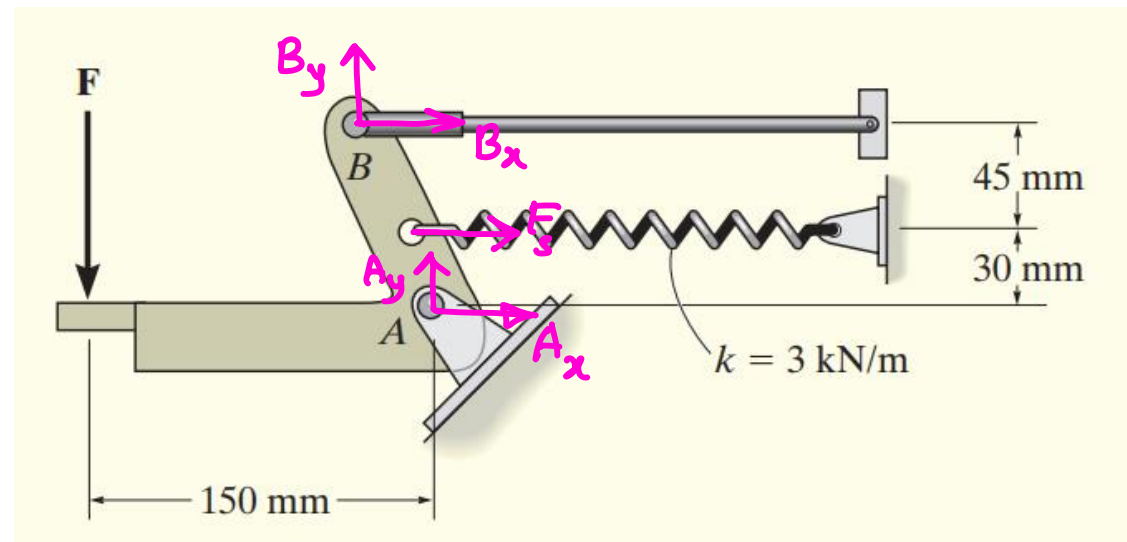
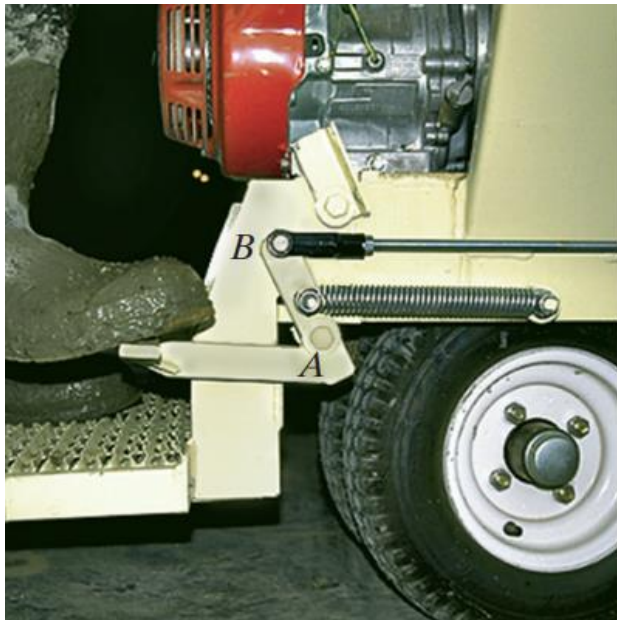
Concluding Notes:

- No equilibrium problem should be solved without *first drawing the free-body diagram*, so as to account for all the forces and couple moments that act on the body.
- If a support *prevents translation* of a body, then the support, when it is removed, exerts a *force* on the body.
- If *rotation is prevented*, then the support, when it is removed, exerts a *couple moment* on the body.
- Internal forces are *never shown* on the free-body diagram since they occur in equal but opposite collinear pairs and therefore cancel out.
- The weight of a body is an external force, and its effect is represented by a single resultant force acting through the body's center of gravity G .
- *Couple moments* can be placed anywhere on the free-body diagram since they are *free vectors*. *Forces* can act at any point along their lines of action since they are *sliding vectors*.

Example: Draw the free-body diagram of the uniform beam shown below. The beam has a mass of 100 kg.



Example: Draw the free-body diagram of the foot lever shown below. The operator applies a vertical force to the pedal so that the spring is stretched 50 mm. and the force on the link at B is 100 N.



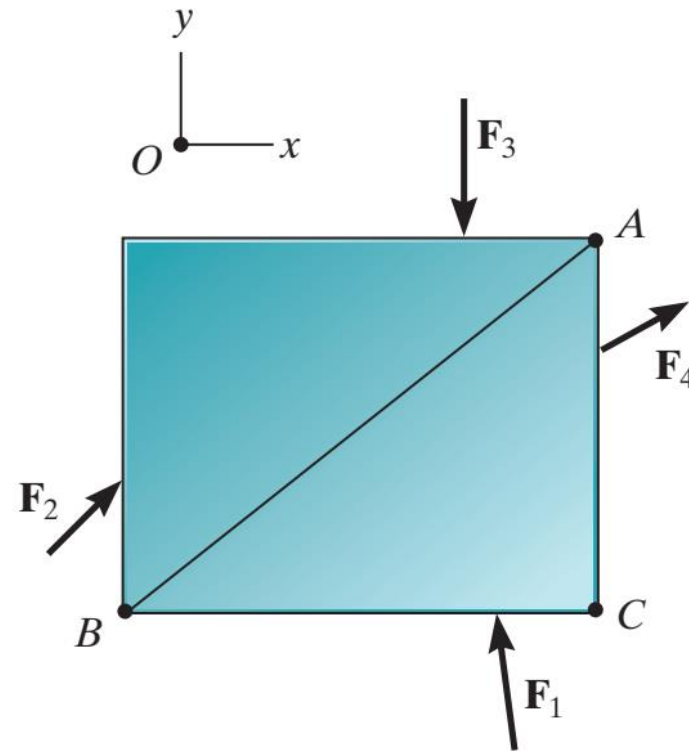
Equilibrium in 2D: Equations of Equilibrium

General equations

$$\Sigma F_x = 0$$

$$\Sigma F_y = 0$$

$$\Sigma M_O = 0$$



Equilibrium in 2D: Equations of Equilibrium

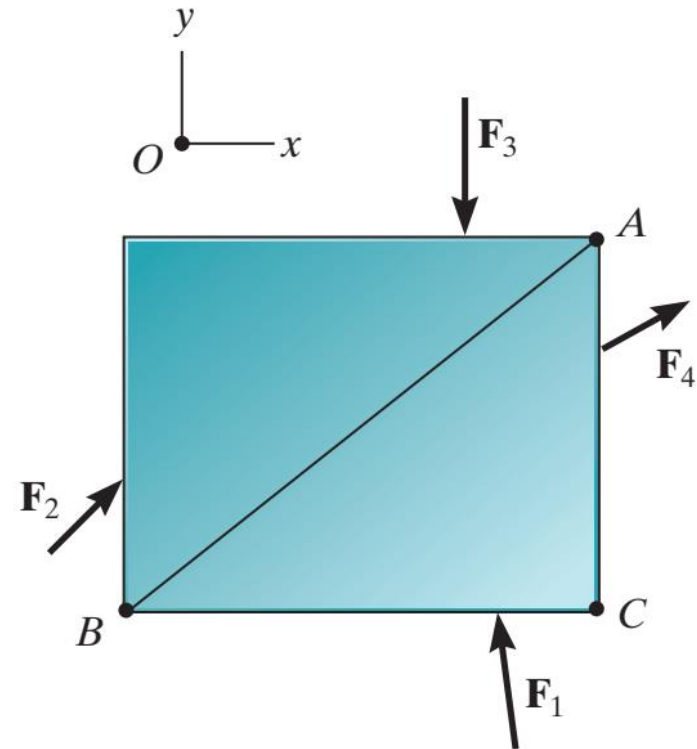
Alternative Sets of Equilibrium Equations

$$\Sigma F_x = 0$$

$$\Sigma M_A = 0$$

$$\Sigma M_B = 0$$

When using these equations, it is required that a line passing through points A and B is **not parallel** to the y axis.



Equilibrium in 2D: Equations of Equilibrium

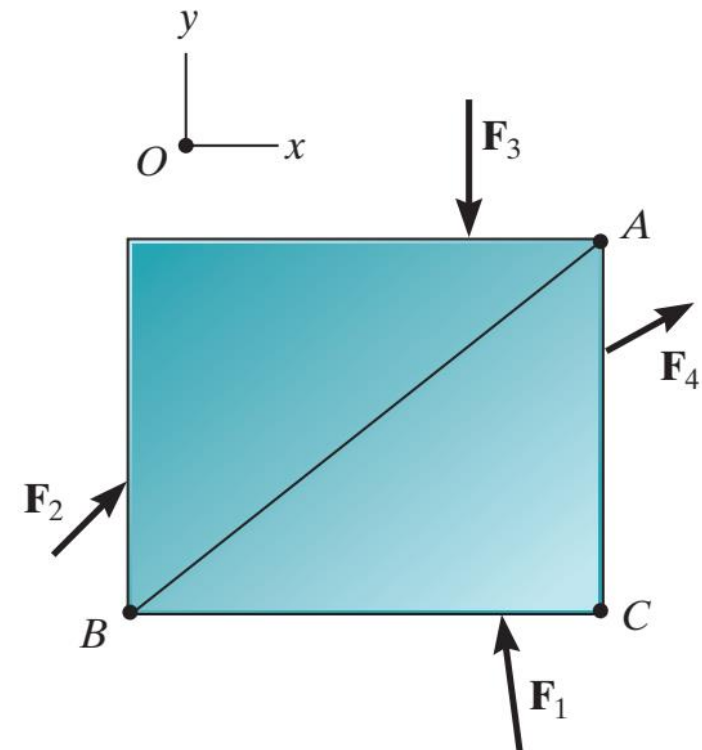
Alternative Sets of Equilibrium Equations

$$\Sigma M_A = 0$$

$$\Sigma M_B = 0$$

$$\Sigma M_C = 0$$

Here it is necessary that points A , B , and C do **not lie on the same line**.



Example: Determine the horizontal and vertical components of reaction on the beam caused by the pin at B and the rocker at A as shown below. Neglect the weight of the beam.

Diagram of a beam with a pin support at A and a roller support at B . The beam is horizontal. A force of 600 N is applied at a 45° angle, 2 m from A . A downward force of 100 N is applied 5 m from A . A downward force of 200 N is applied 7 m from A . The distance between A and B is 7 m ($2\text{ m} + 3\text{ m} + 2\text{ m}$). The reaction at A is A_y (up). The reaction at B is B_x (left) and B_y (up).

Handwritten calculations:

$$\sum F_y = 0$$

$$A_y - 600 \sin 45^\circ - 200 - 100 + B_y = 0$$

$$B_y = 404.75\text{ N}$$

$$\sum F_x = 0 \Rightarrow B_x = -600 \cos 45^\circ$$

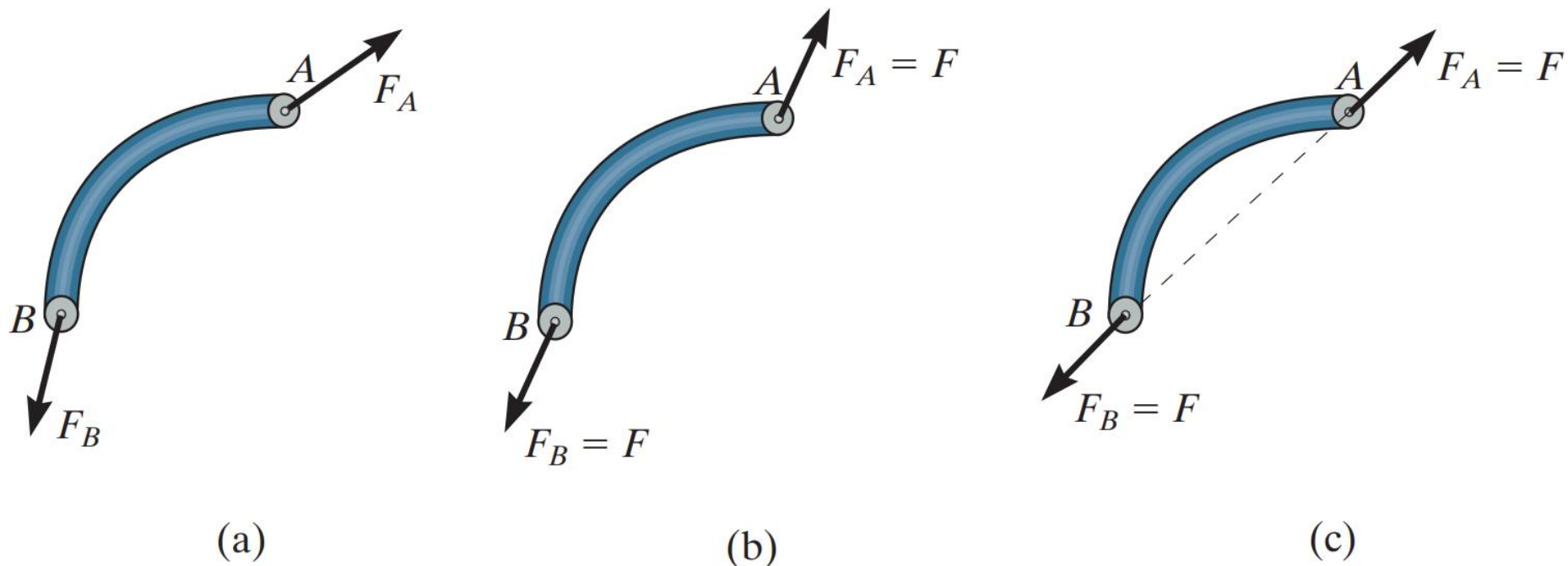
$$\sum M_B = 0$$

$$-A_y(7) + 600 \sin 45^\circ(5) - 600 \cos 45^\circ(0.2) + 100(2) = 0$$

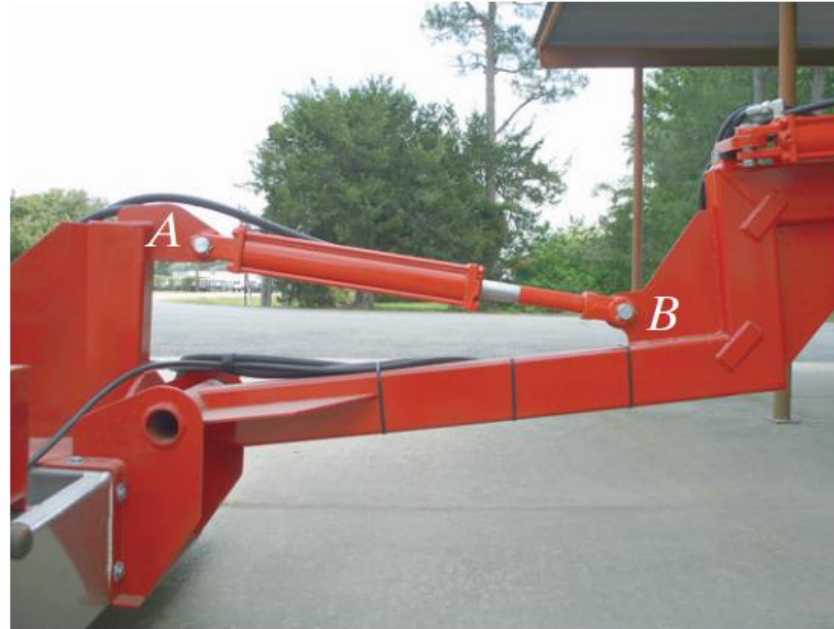
$$A_y = 319.5\text{ N}$$

Two-Force Members

As the name implies, a ***two-force member*** has forces applied at only two points on the member.



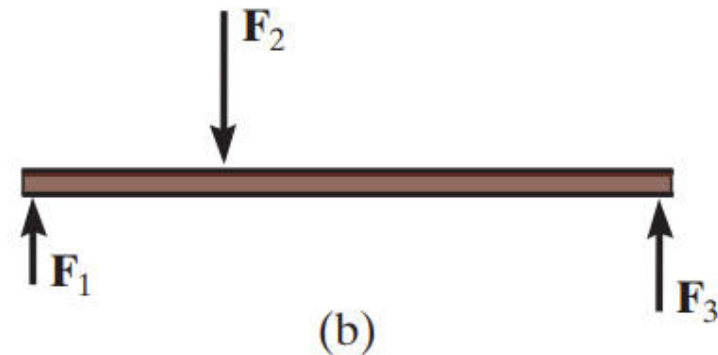
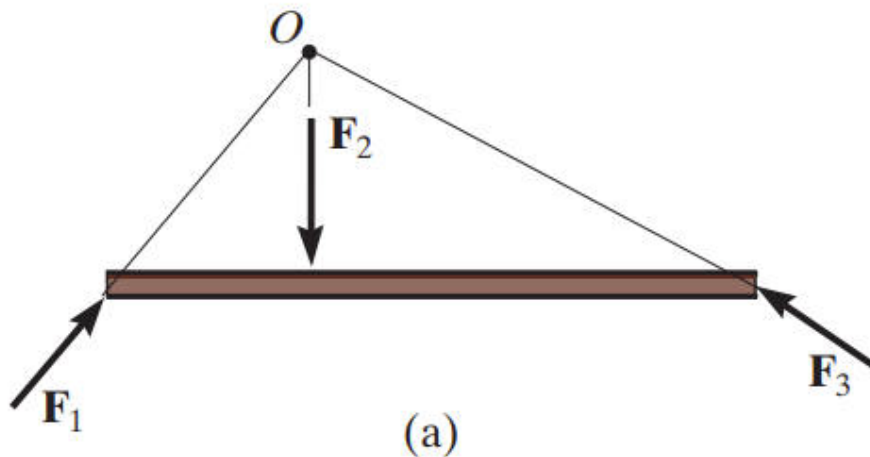
Two-Force Members



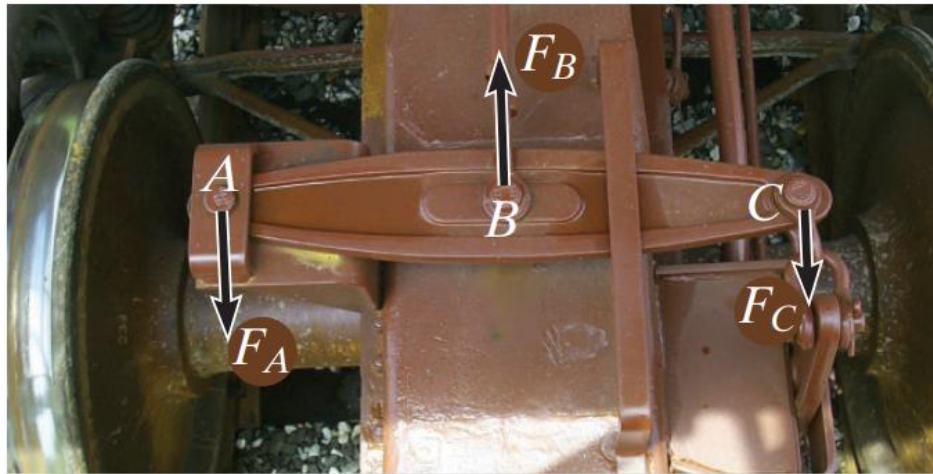
The hydraulic cylinder AB is a typical example of a two-force member since it is pin connected at its ends and, provided its weight is neglected, only the resultant pin forces act on this member.

Three-Force Members

If a member is subjected to only *three forces*, it is called a ***three-force member***. Moment equilibrium can be satisfied only if the three forces form a *concurrent* or *parallel* force system.



Three-Force Members



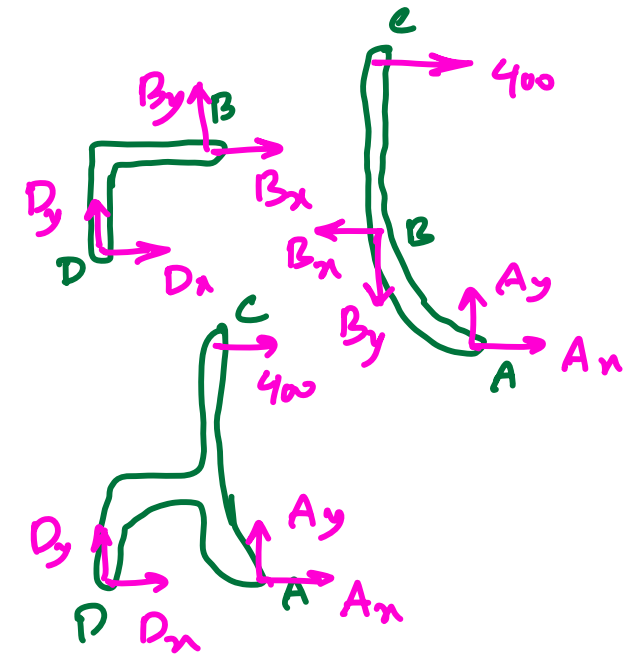
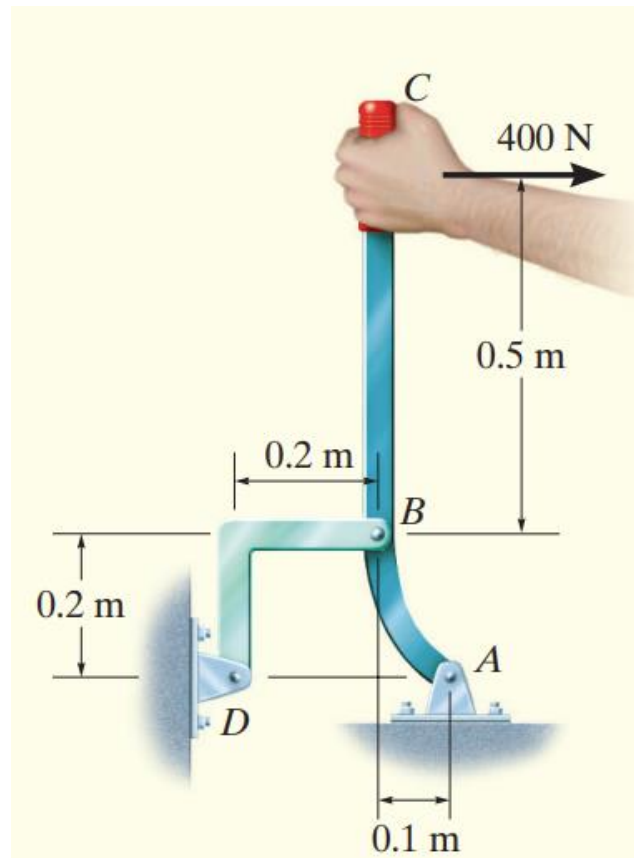
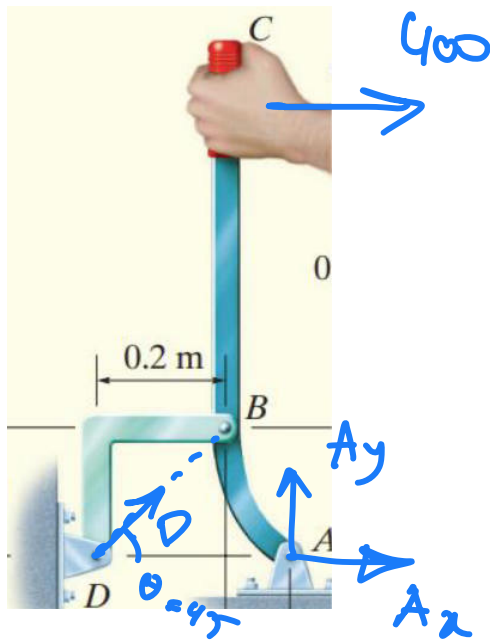
The link used for this railroad car brake is a three-force member. Since the force $\mathbf{F}_B$ in the tie rod at B and $\mathbf{F}_C$ from the link at C are parallel, then for equilibrium the resultant force $\mathbf{F}_A$ at the pin A must also be parallel with these two forces.

Three-Force Members



The boom and bucket on this lift is a three-force member, provided its weight is neglected. Here the lines of action of the weight of the worker, $\mathbf{W}$, and the force of the two-force member (hydraulic cylinder) at B , $\mathbf{F}_B$, intersect at O . For moment equilibrium, the resultant force at the pin A , $\mathbf{F}_A$, must also be directed towards O .

Example: The lever ABC is pin supported at A and connected to a short link BD as shown below. If the weight of the members is negligible, determine the force of the pin on the lever at A .



Practice at Home

1. Two smooth pipes, each having a mass of 300 kg, are supported by the forked tines of the tractor shown below. Draw the free-body diagrams for each pipe and both pipes together.



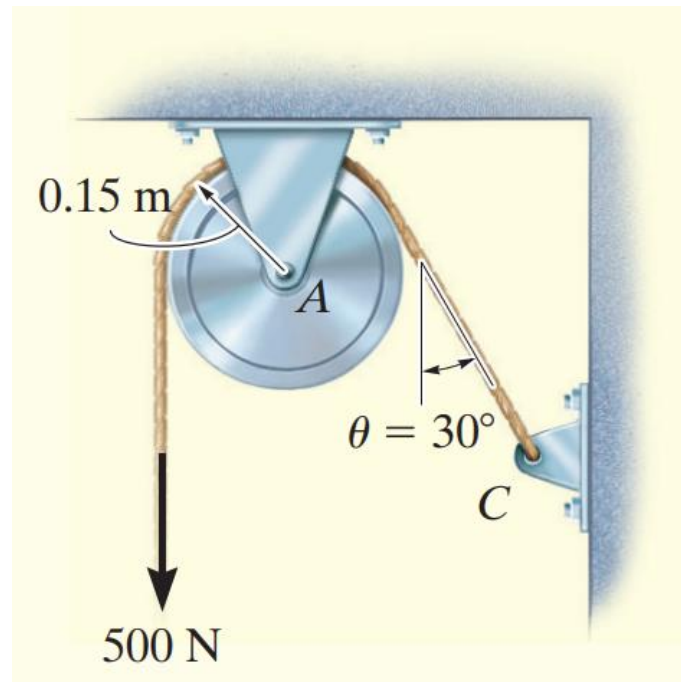
Practice at Home

2. Draw the free-body diagram of the unloaded platform that is suspended off the edge of the oil rig shown below. The platform has a mass of 200 kg.



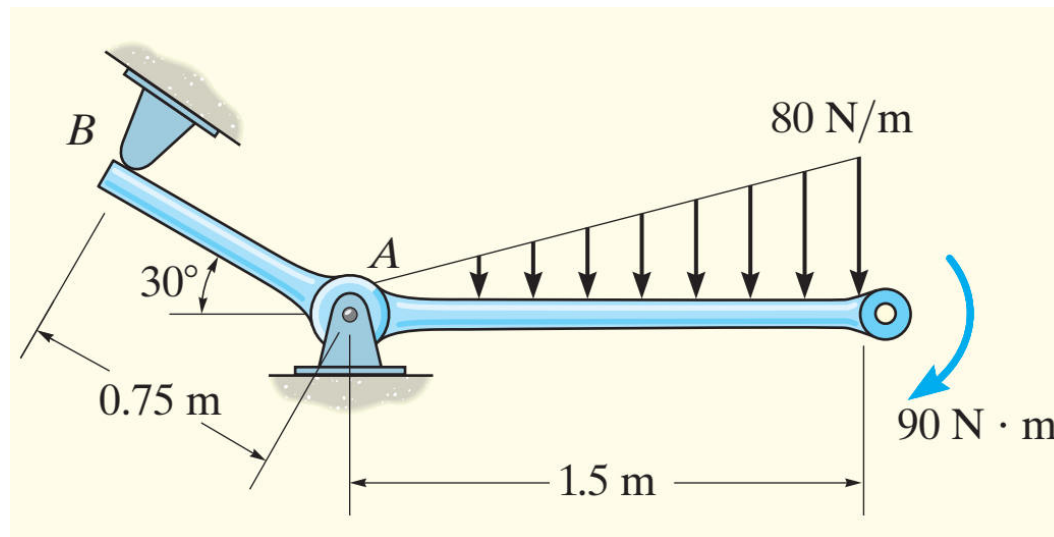
Practice at Home

3. The cord shown below supports a force of 500 N and wraps over the frictionless pulley. Determine the tension in the cord at C and the horizontal and vertical components of reaction at pin A.



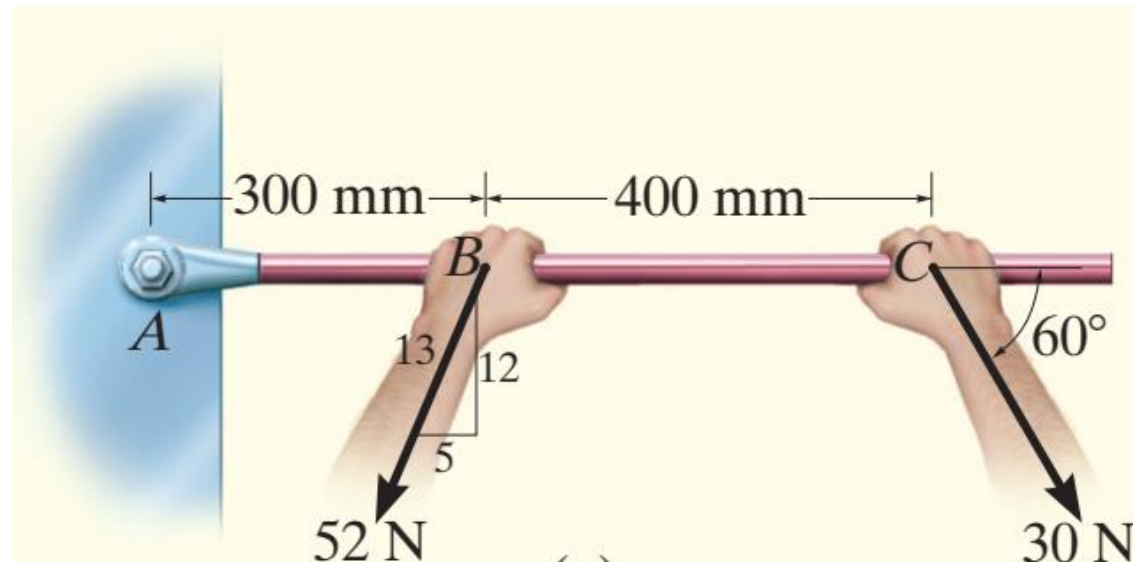
Practice at Home

4. The member shown below is pin connected at A and rests against a smooth support at B . Determine the horizontal and vertical components of reaction at pin A .



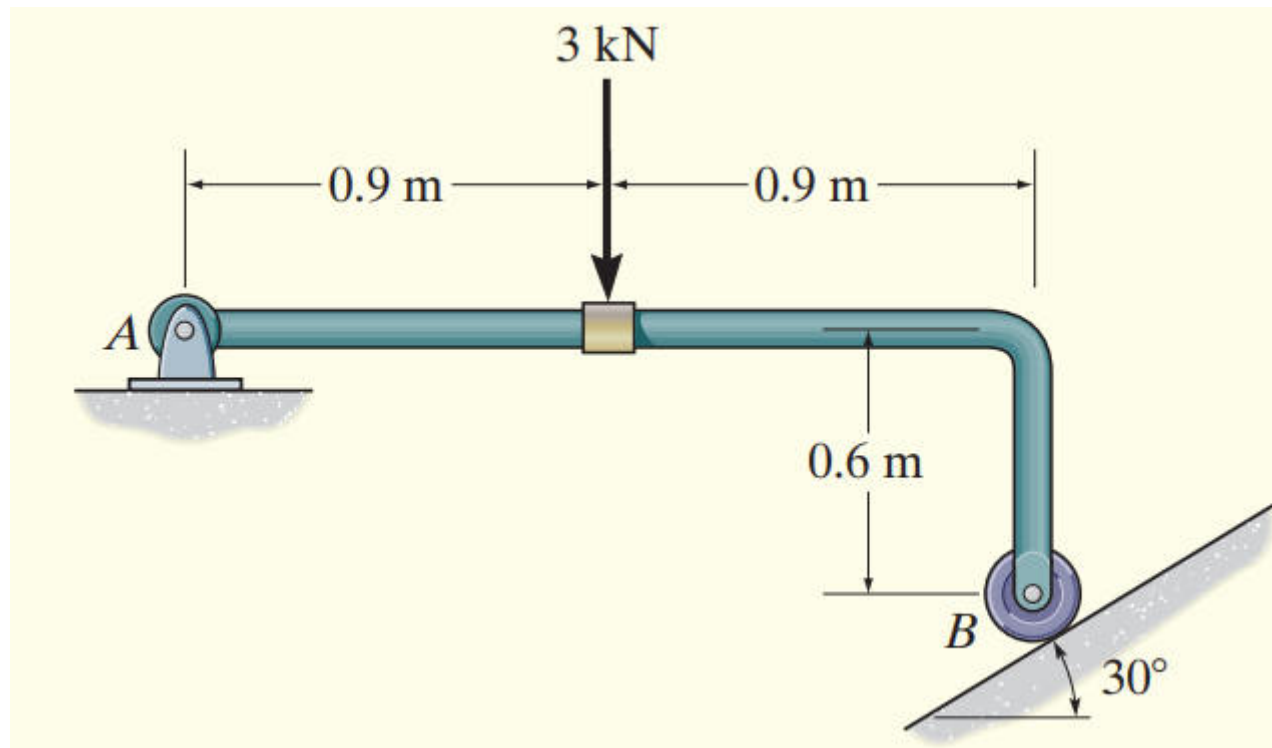
Practice at Home

5. The box wrench shown below is used to tighten the bolt at A . If the wrench does not turn when the load is applied to the handle, determine the torque or moment applied to the bolt and the force of the wrench on the bolt.



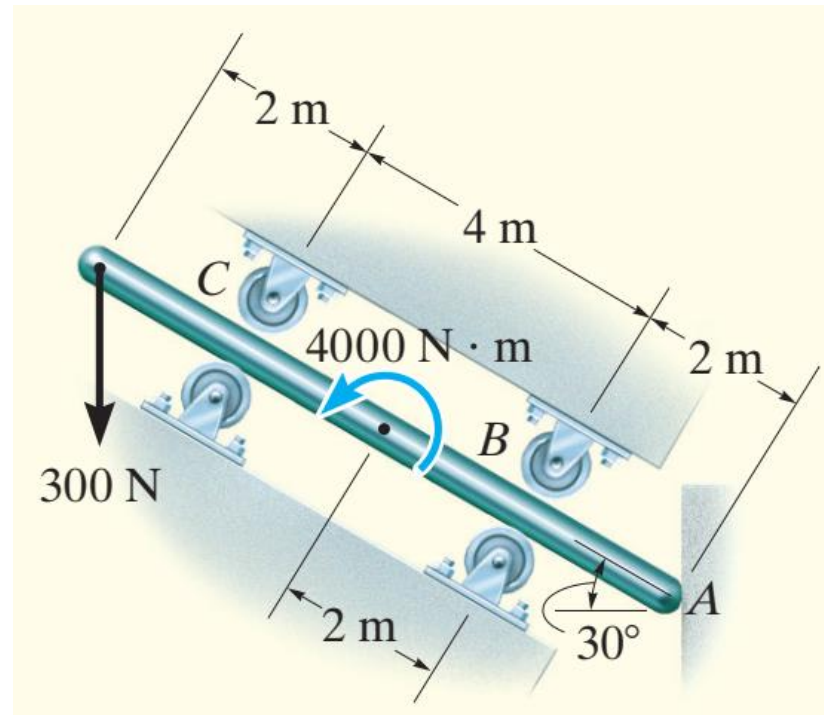
Practice at Home

6. Determine the horizontal and vertical components of reaction on the member at pin A , and the normal reaction at the roller B shown in the figure below.



Practice at Home

7. The uniform smooth rod shown below is subjected to a force and couple moment. If the rod is supported at *A* by a smooth wall and at *B* and *C* either at the top or bottom by rollers, determine the reactions at these supports. Neglect the weight of the rod.



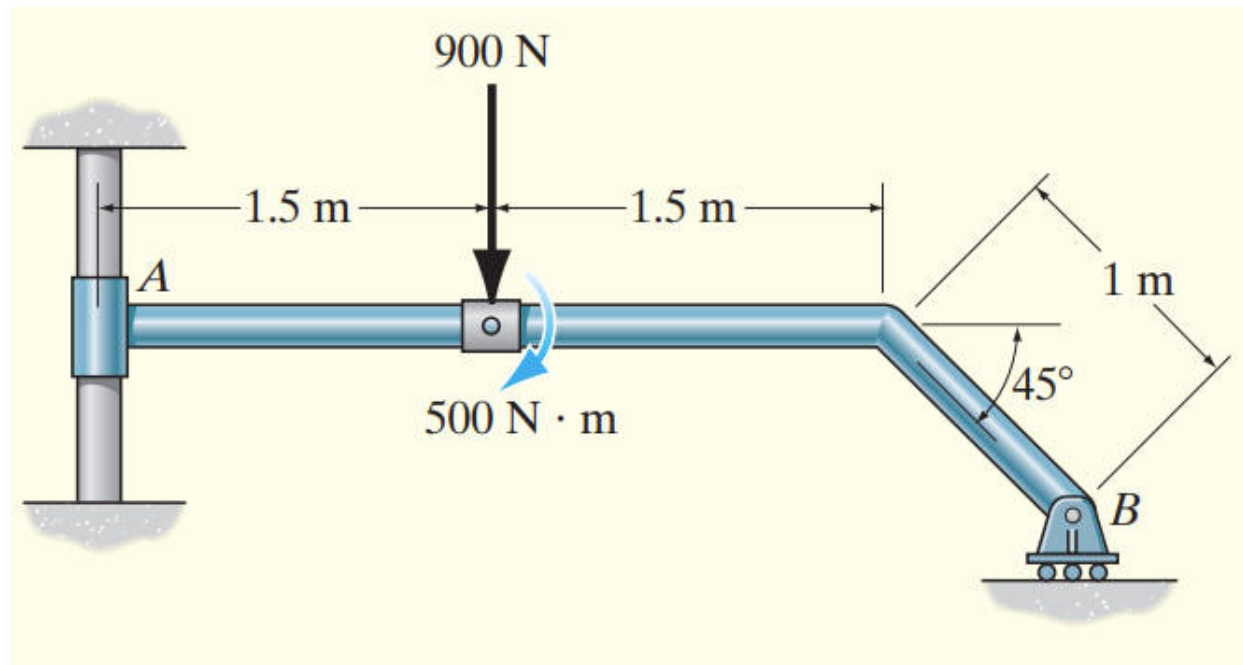
Practice at Home

8. The uniform truck ramp shown below has a weight of 2000 N and is pinned to the body of the truck at each side and held in the position shown by the two side cables. Determine the tension in the cables.



Practice at Home

9. Determine the support reactions on the member below. The collar at A is fixed to the member and can slide vertically along the vertical shaft.



Practice at Home

10. The uniform bar OC of length L pivots freely about a horizontal axis through O . If the spring of modulus k is unstretched when C coincides with A , determine the tension T required to hold the bar in the position shown. The diameter of the small pulley at D is negligible.

